

Architecture Document – Deployment and Operations

Deployment Topology

Services run on a Kubernetes cluster across 3 availability zones, with at least 2 replicas per service and anti-affinity rules.

Observability

The observability stack uses Prometheus for metrics, Loki for logs and Tempo for traces, correlated by trace_id.

Backup Policy

Databases are backed up in full daily with 35-day retention, incrementally every 6 hours, and restore drills run quarterly.

Availability Target

The availability target for core services is 99.95%, corresponding to a monthly unavailability budget of about 21 minutes.

Capacity and Scaling

Capacity is provisioned at twice peak traffic. Stateless services scale horizontally; stateful components scale through sharding or read-write splitting.

Security Design

The architecture follows least privilege and zero trust. Service-to-service calls require authentication and authorisation, and secrets are delivered through a key management service.

ADR

Architecture decisions are recorded as ADRs, and material changes require Architecture Review Board approval. This document describes the target architecture; the configuration repository is authoritative.

Design Goals

The platform architecture targets high availability, scalability and observability, prioritising data consistency and fault isolation on core transactional paths.